Circuit investigation

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use measurements to test a circuit model safely.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 7.*
12 V; drops add to the supply voltage.

02. Worked example: explain the model
Straight line slope 10 ohms; behavior beyond tested conditions remains untested.

03. Read and interpret the model
Ammeter in series; voltmeter parallel across resistor.

04. Apply the physics
0.50 A predicted. Actual values must be measured, not copied from prediction.

05. Explain your reasoning
Check connections, meter range/placement, units, component values, supply voltage; retain raw evidence.

06. Review: connect an earlier idea *Review: Unit 6, Day 16.*
It becomes 1/4 as large by Wien’s law.

07. Exit check
Teacher verifies connections and safety before power; student records actual feedback.

08. STRETCH: test the model
Expected 0.8 A; not a fit. Check heating, contacts, meters, component identity and uncertainty.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 7.*
New resistance (3+2) R=5 R; current ratio 3/5≈0.6.

02. Worked example: explain the model
Straight line slope 10 ohms; behavior beyond tested conditions remains untested.

03. Read and interpret the model
Ammeter in series; voltmeter parallel across resistor.

04. Apply the physics
12.61 Ω; 5.07% above nominal. Meter uncertainty, resistor tolerance or heating may contribute.

05. Explain your reasoning
Check connections, meter range/placement, units, component values, supply voltage; retain raw evidence.

06. Review: connect an earlier idea *Review: Unit 6, Day 16.*
T≈5000 K (about 5000 K); peak wavelength is rounded.

07. Exit check
Vary safe supply voltage; measure current and resistor voltage; control resistor/temperature; constant V/I or linear V-I through origin.

08. STRETCH: test the model
Central 10 ohms; minimum 2.9/0.32≈9.06, maximum 3.1/0.28≈11.07 ohms.